

Class Notes: Phylum Coelenterata (Cnidaria)

Animal Kingdom • PU-I Biology • Class Notes

Organisation and Cnidoblasts

Coelenterates/Cnidaria are aquatic (mostly marine), sessile or free-swimming, radially symmetrical, diploblastic, with tissue level of organisation. Have a central gastro-vascular cavity (GVC) with a single opening, the hypostome, surrounded by tentacles.

Forms, Digestion and Reproduction

Tentacles have cnidoblasts/cnidocytes containing nematocysts, used for anchorage, defence and capturing prey — the name Cnidaria comes from cnidoblasts. Digestion can be extracellular and intracellular; skeleton (in corals) is composed of calcium carbonate. Two basic body forms: polyp (sessile, cylindrical, e.g. Hydra, Adamsia) and medusa (umbrella-shaped, free-swimming, e.g. Aurelia/jellyfish).

Examples

Cnidarians showing both forms exhibit alternation of generation (metagenesis): polyps produce medusae asexually, medusae form polyps sexually (e.g. Obelia). Examples: Physalia (Portuguese man of war), Adamsia (sea anemone), Pennatula (sea-pen), Gorgonia (sea-fan), Meandrina (brain coral), Obelia (sea fur), Aurelia (jellyfish).